How Does an RC Circuit Charge and Discharge?

An RC circuit is a fundamental electrical circuit consisting of a resistor and a capacitor connected in series or parallel. RC circuits are crucial for understanding charging and discharging behaviors of capacitors, which are important in electronic devices and JEE Main Physics. This article provides a clear explanation of RC circuit concepts, equations, analysis methods, and solved examples for JEE Main.

Basic Structure and Function of an RC Circuit

An RC circuit contains a resistor (R) and a capacitor (C) connected in a specific arrangement. When a voltage source is applied, the circuit exhibits time-dependent behavior as the capacitor charges or discharges through the resistor.

The resistor controls the flow of electric current, while the capacitor stores electric charge. The rate of charging or discharging in the circuit is governed by both $R$and Cvalues. For foundational concepts, refer to RC Circuit Overview.

Charging of a Capacitor in an RC Circuit

When a capacitor is connected in series with a resistor and a DC voltage source, it starts charging. The charging process is exponential and does not occur instantly. The current and charge change with time until the capacitor is fully charged.

The charge on the capacitor at any time $t$is given by the equation:

$$q = q_0[1 - e^{-t/RC}]$$

Here, $q_0 = C\varepsilon$is the maximum charge, where $\varepsilon$is the applied emf. The current in the circuit at any time $t$is:

$$I = I_0 e^{-t/RC}$$

$I_0$is the initial current, given by $I_0 = \frac{\varepsilon}{R}$. The charging is considered nearly complete after a time interval much greater than RC.

RC Time Constant and Its Importance

The product $RC$is known as the time constant, denoted by τ. It determines the rate at which the capacitor charges or discharges. The time constant is measured in seconds and is a key parameter in RC circuits.

After a time interval equal to one time constant $(t = RC)$, the charge on the capacitor reaches about 63% of its maximum value. The current falls to approximately 37% of its initial value.

Parameter	Value at $t = RC$
Charge on Capacitor (q)	$0.63q_0$
Current (I)	$0.37I_0$

Derivation of Charging Formulas Using Kirchhoff's Law

Kirchhoff's Voltage Law is applied to analyze the RC circuit during charging. The law states that the sum of voltages in a closed loop equals zero. For a series RC circuit, the equation is written as:

$$\varepsilon - iR - \frac{q}{C} = 0$$

Rearranging and integrating this differential equation yields the standard charging equation for both charge and current in the circuit. For a detailed understanding of capacitors' role, refer to Explore Capacitors.

Discharging of a Capacitor in an RC Circuit

When the charged capacitor is disconnected from the emf source and allowed to discharge through the resistor, the process also follows an exponential law. The current and charge decrease over time as the capacitor releases stored energy.

The charge remaining on the capacitor at time $t$during discharge is:

$$q = q_0 e^{-t/RC}$$

The current in the discharging circuit is:

$$I = -I_0 e^{-t/RC}$$

The negative sign indicates that the current direction is opposite to that of the charging current. For more on electric current flow, see Electric Current Fundamentals.

Voltage Across Resistor and Capacitor

The voltage across the resistor and capacitor varies with time. During charging:

- Voltage across capacitor: $V_C = \varepsilon(1 - e^{-t/RC})$
- Voltage across resistor: $V_R = \varepsilon e^{-t/RC}$

During discharging, both voltages decrease exponentially, following the respective exponential equations for $q$and I. For more on resistance behavior, refer to Ohm's Law and Resistance.

Energy Considerations and Heat Dissipation

The energy supplied by the battery in the RC circuit is partly stored in the capacitor and partly dissipated as heat in the resistor. The heat dissipated equals $\frac{1}{2}C\varepsilon^2$after a long time for a complete charging process.

Energy Form	Expression
Capacitor Stored Energy	$\frac{1}{2}C\varepsilon^2$
Heat Dissipated	$\frac{1}{2}C\varepsilon^2$

Time Constant Calculation in Complex RC Circuits

In circuits with multiple resistors and capacitors, the time constant may be calculated using Thevenin's theorem. This involves reducing the circuit to a single equivalent resistor and using the same capacitor. The effective time constant is $\tau = R_{\text{eff}}C$.

To analyze such circuits, first remove the capacitor and calculate the resistance seen across its terminals, then multiply by the capacitance to determine τ. For circuit simplification methods, refer to Circuit Solving Techniques.

Analysis of Complex RC Networks

For RC circuits with multiple branches, Kirchhoff's Laws are employed to write equations for each loop. These equations are then solved to find the time-dependent charge and current in each branch. For advanced RC circuit arrangements, combine resistors using series and parallel rules before applying the RC equations.

RC Circuit Equations and Formulas Summary

The following key equations are used in RC circuit analysis:

- **Charge during charging:** $q = C\varepsilon[1 - e^{-t/RC}]$
- **Current during charging:** $I = \frac{\varepsilon}{R}e^{-t/RC}$
- **Charge during discharging:** $q = q_0 e^{-t/RC}$
- **Current during discharging:** $I = -\frac{q_0}{RC}e^{-t/RC}$
- **Time constant:** $\tau = RC$

Solved Examples: RC Circuits

1. A $10\ \mu$Fcapacitor is charged through a 2 kΩresistor by a 20 Vbattery. Calculate the charge after one time constant.

Time constant, $\tau = RC = (2 \times 10^3)(10 \times 10^{-6}) = 0.02$ s. Maximum charge, $q_0 = C\varepsilon = 10 \times 10^{-6} \times 20 = 2 \times 10^{-4}$ C. After one τ, $q = q_0(1 - e^{-1}) \approx 0.63 q_0 = 1.26 \times 10^{-4}$ C.

2. A capacitor of $5\ \mu$Fwith initial charge $15\ \mu$Cis discharged through a $10\ \Omega$resistor. Find the remaining charge after $50\ \mu$s.

$\tau = RC = 10 \times 5 \times 10^{-6} = 50\ \mu$s. After $t = \tau$, $q = q_0 e^{-1} \approx 15\ \mu\text{C} \times 0.37 = 5.55\ \mu$C.

RC Circuit Applications and Further Study

RC circuits are widely used in timing devices, filtering operations, and analog circuits. Understanding RC circuit analysis is essential for physics problem-solving and lays the foundation for topics involving AC circuits. For topics related to inductor behavior, see [Understanding Inductors]